

Simulation of mosquito-borne disease spread in Denmark

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Objective

Assess how vulnerable Denmark would be under a newly introduced mosquito-borne pathogen

How?

Building a generic model for spatial mosquito-borne disease spread. It needs:

- A flexible host structure (livestock and wildlife, e.g. wild birds)
- Multiple transmission pathways (mosquito-borne, directly host-to-host, between-farms)

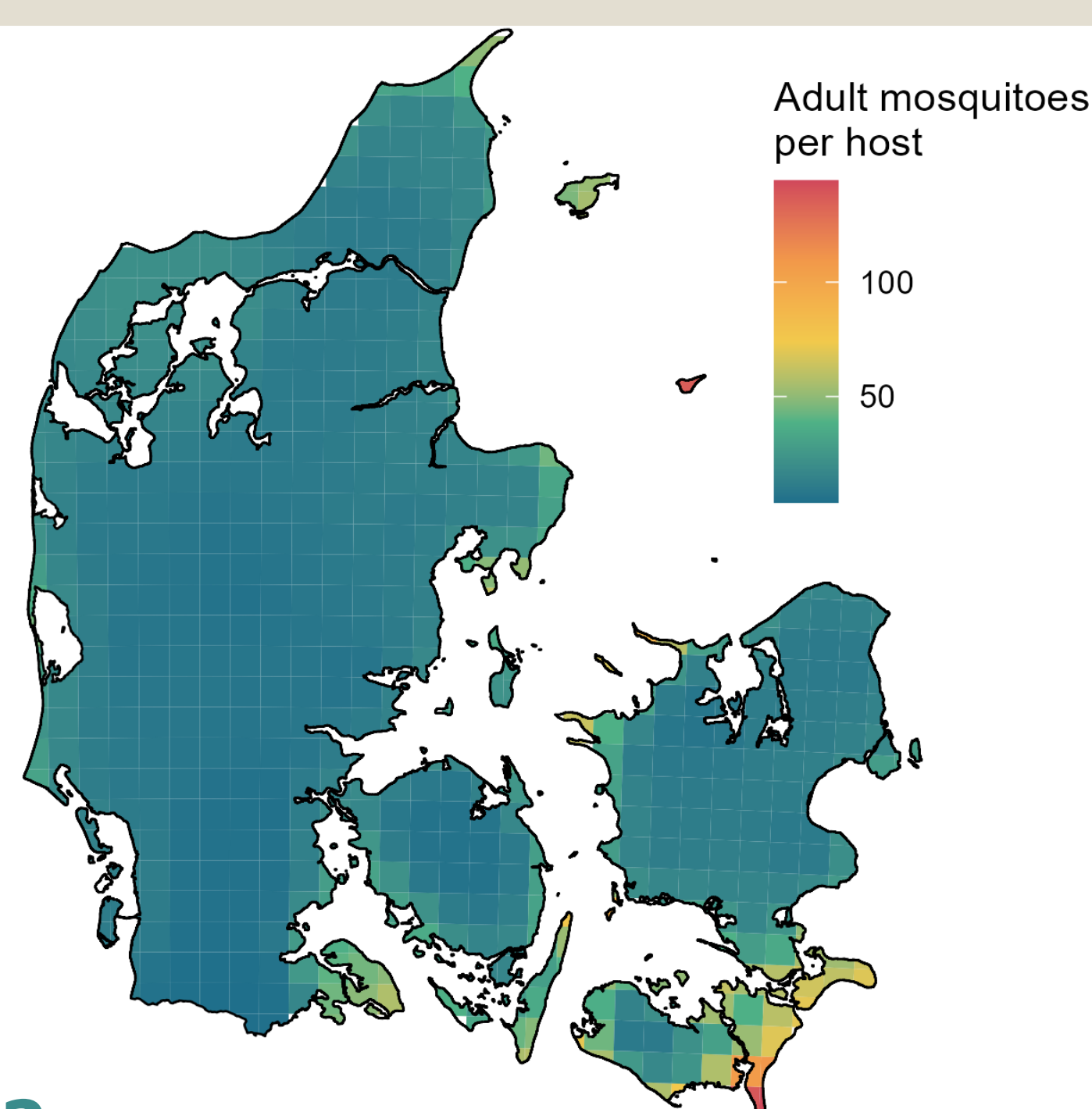
We want to be able to test control measures and identify main drivers of transmission

What we already have

1. Mosquito abundance predictions

In a parallel project, we have analysed mosquito abundance data to obtain **predictions of mosquito abundance** depending on temperature (See Figure 1).

Figure 1. Number of Adult mosquitoes per host on the 2nd of September 2024. Predicted by a population dynamic model with temperature dependence. Temperature has been obtained in a 10 x 10 km grid cell resolution (1).



2. Farm data

We have access to Danish Husbandry Registry:

- **Location** of farms
- **Number of animals (by species)** on each farm (over time)

→ In progress: Acquiring **animal movement data**

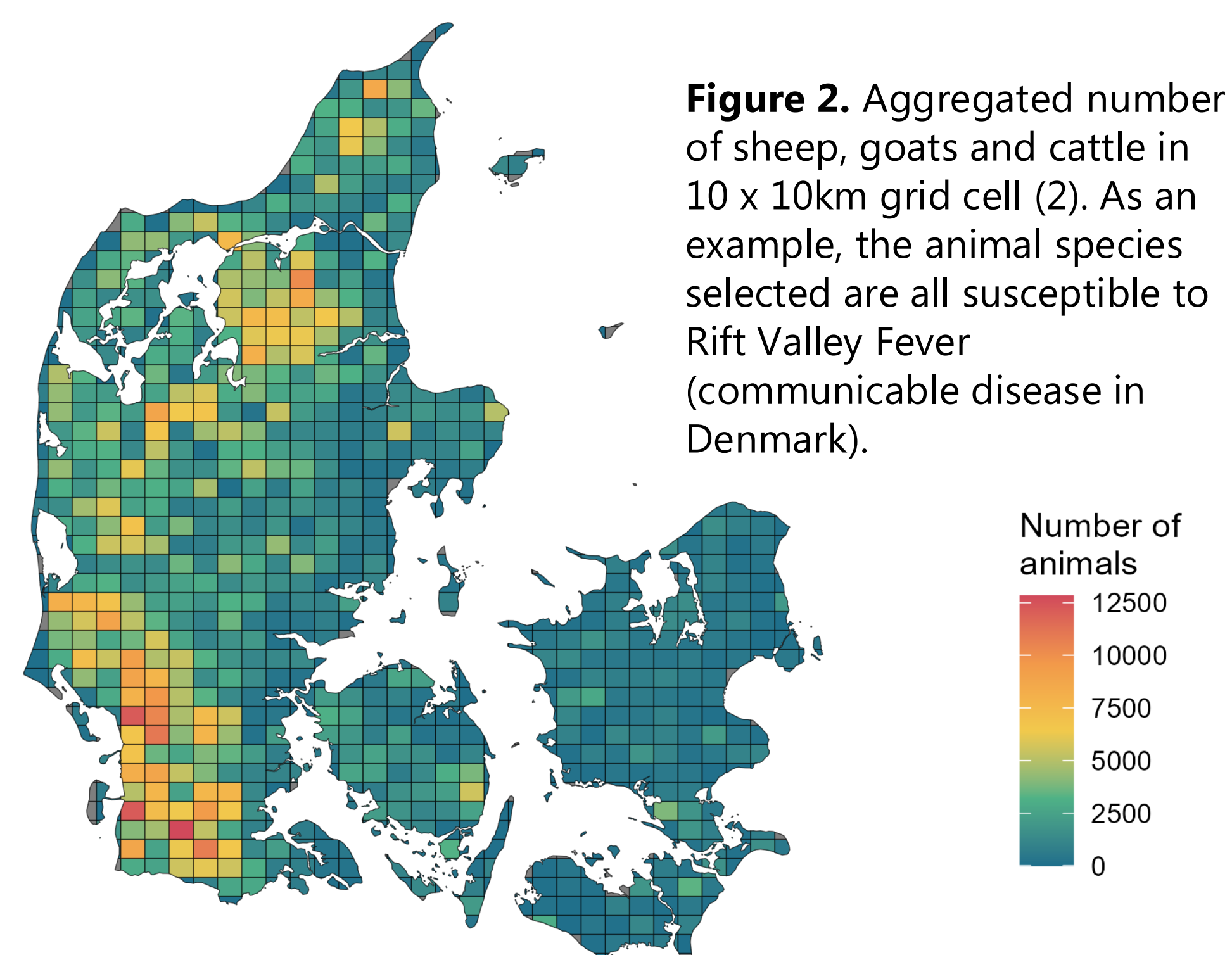


Figure 2. Aggregated number of sheep, goats and cattle in 10 x 10km grid cell (2). As an example, the animal species selected are all susceptible to Rift Valley Fever (communicable disease in Denmark).

Modelling the hosts

Compartmental mechanistic model

Host states:

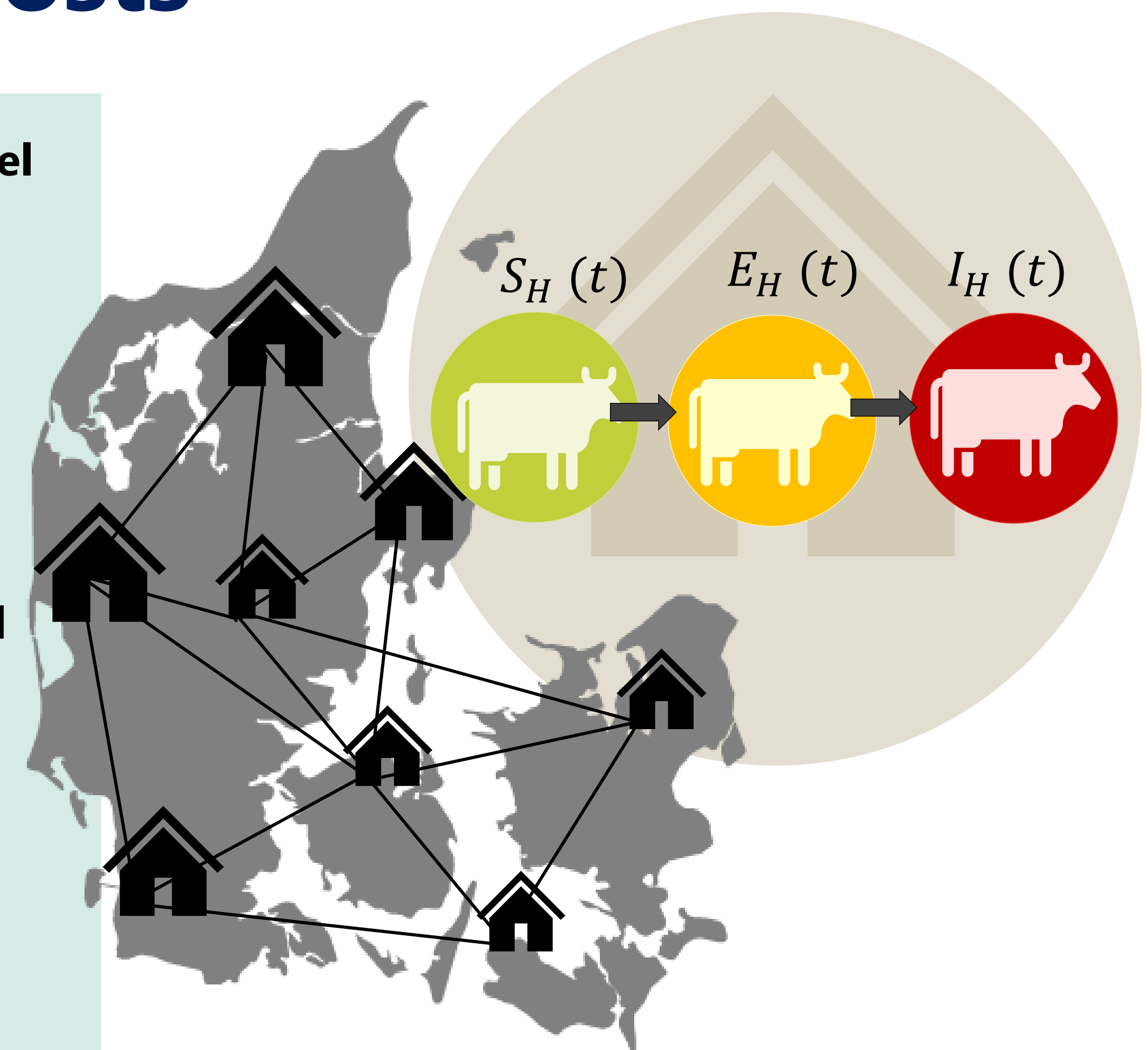
- Susceptible
- Exposed
- Infected
- Recovered

Model features:

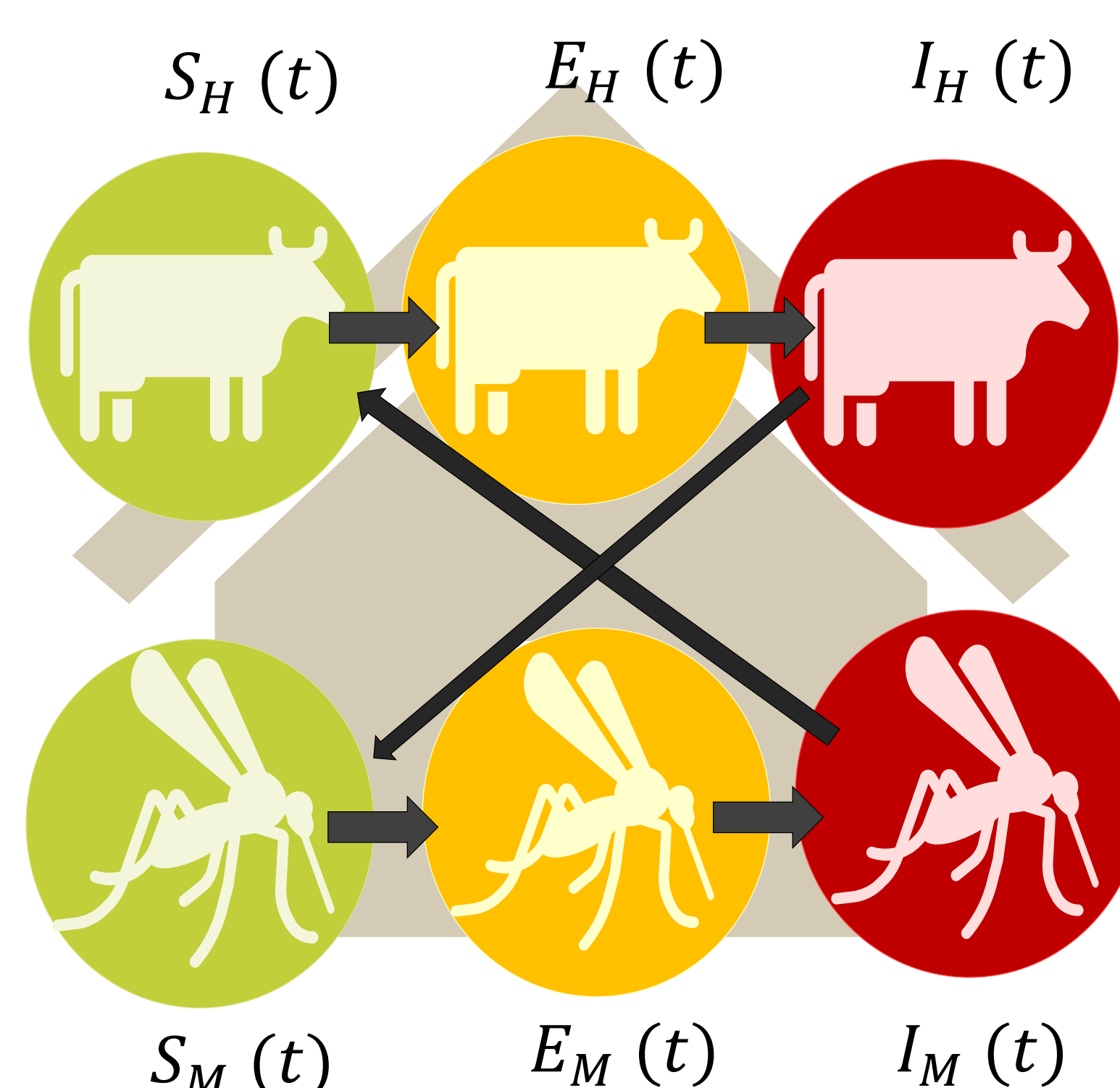
- Species-specific compartments
- Stochastic: Accounts for individual variability among hosts
- Uses animal movement data to simulate between-farm transmission

Advantages:

- Can incorporate between-host transmission
- Control measures (e.g., vaccination) are easily applied



Modelling pathogen spread via mosquitoes – *In progress*



Compartmental mechanistic model

Mosquito states:

- Susceptible
- Exposed
- Infected

Structure: A mosquito population will be assigned to each farm

Advantages:

- Captures biological realism
- Allows easy implementation of mosquito control strategies
- Simple within-farm dynamics

Open questions

How should **mosquito movement between farms** be represented?

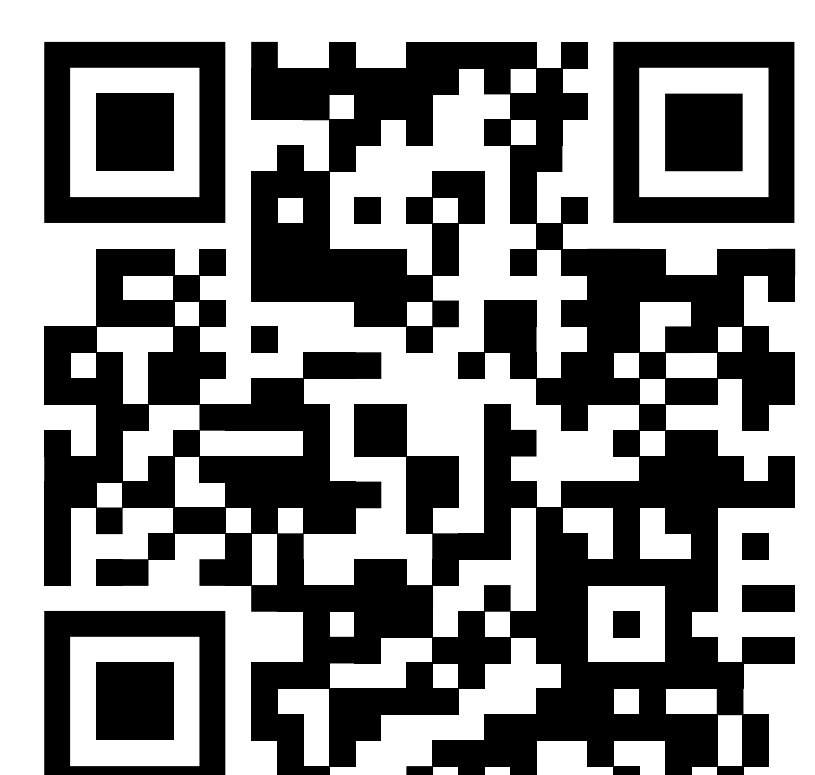
- Wind-driven
- Distance kernel + random jumps between farms

Is it optimal to model **explicit pathogen transmission** through mosquitoes?

- Should pathogen diffuse separately from mosquito population?

This poster shows planned modelling directions. Your input is welcome!

A digital copy of this poster →



Data Sources

(1) Climate Data API, Danish Meteorological Institute (DMI)

(2) Danish Husbandry Registry (VET-MYN Project P1.11)

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